Concept Concentration (COCA)

for Faithful Representation Intervention

Hongzheng Yang [*1] Yongqiang Chen [*2 3] Zeyu Qin [4] Tongliang Liu [5] Chaowei Xiao [6] Kun Zhang [2 3] Bo Han [1]

ICML 2026

What is Representation Intervention?

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- **Localization**

What is Representation Intervention?

- **Localization** : use probing classifiers / activation analysis to find where and how a specific concept is encoded inside the model's hidden layers.

What is Representation Intervention?

- **Localization**

  +

- **Intervention**

What is Representation Intervention?

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- **Intervention** : during inference or training, specific internal activations are altered, clamped, or shifted along a directional vector.

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- **Behavioral Response**

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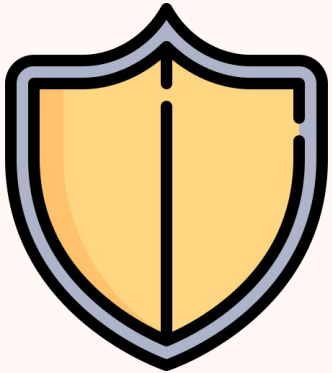


Safety alignment

Why are we tampering with internal representations?



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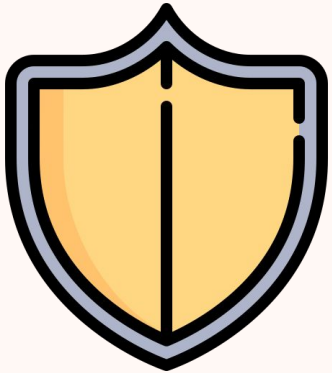


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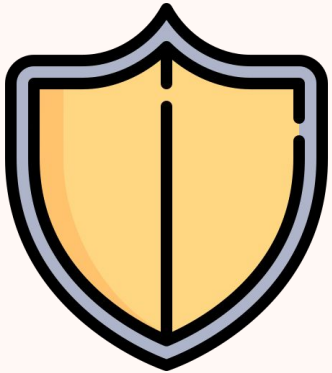


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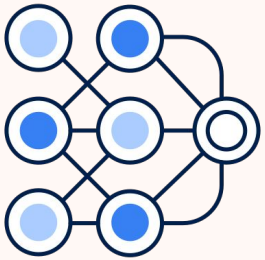
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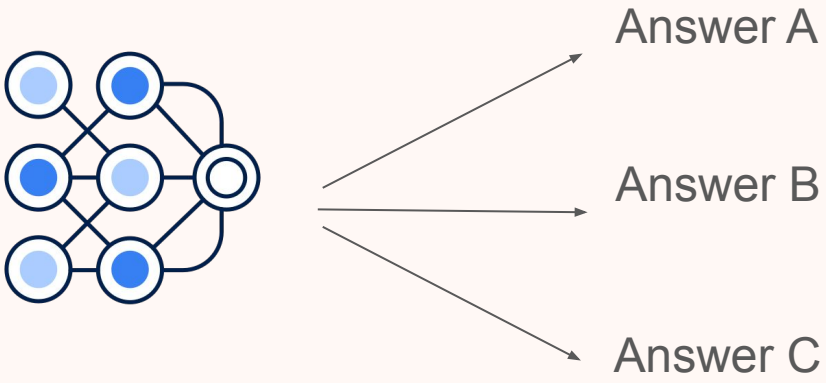
Knowledge editing

Instead of traditional **RLHF** or **Supervised Fine-Tuning**?

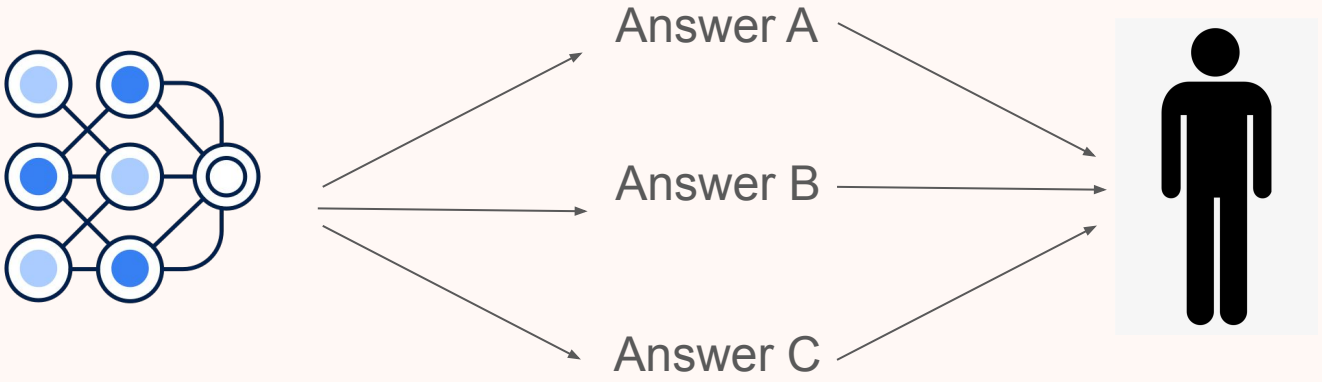
Reinforcement learning from human feedback (**RLHF**)



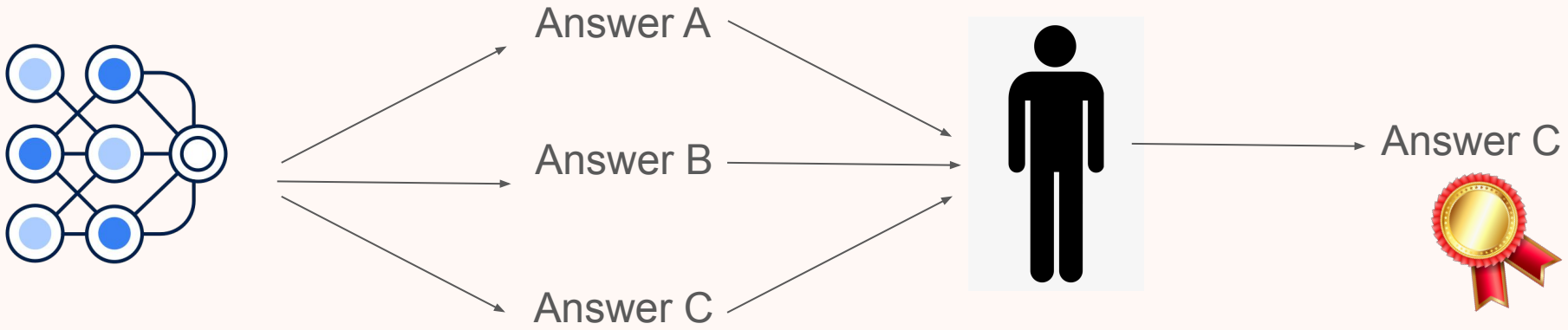
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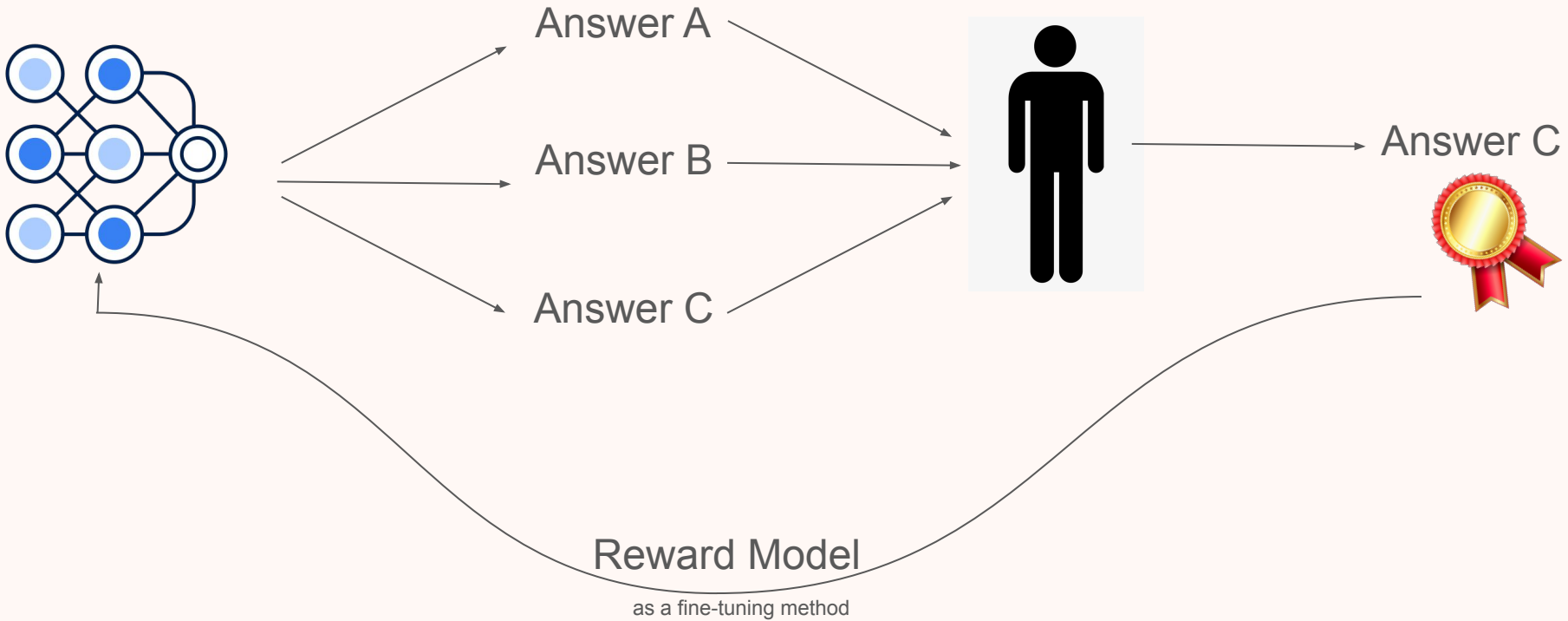
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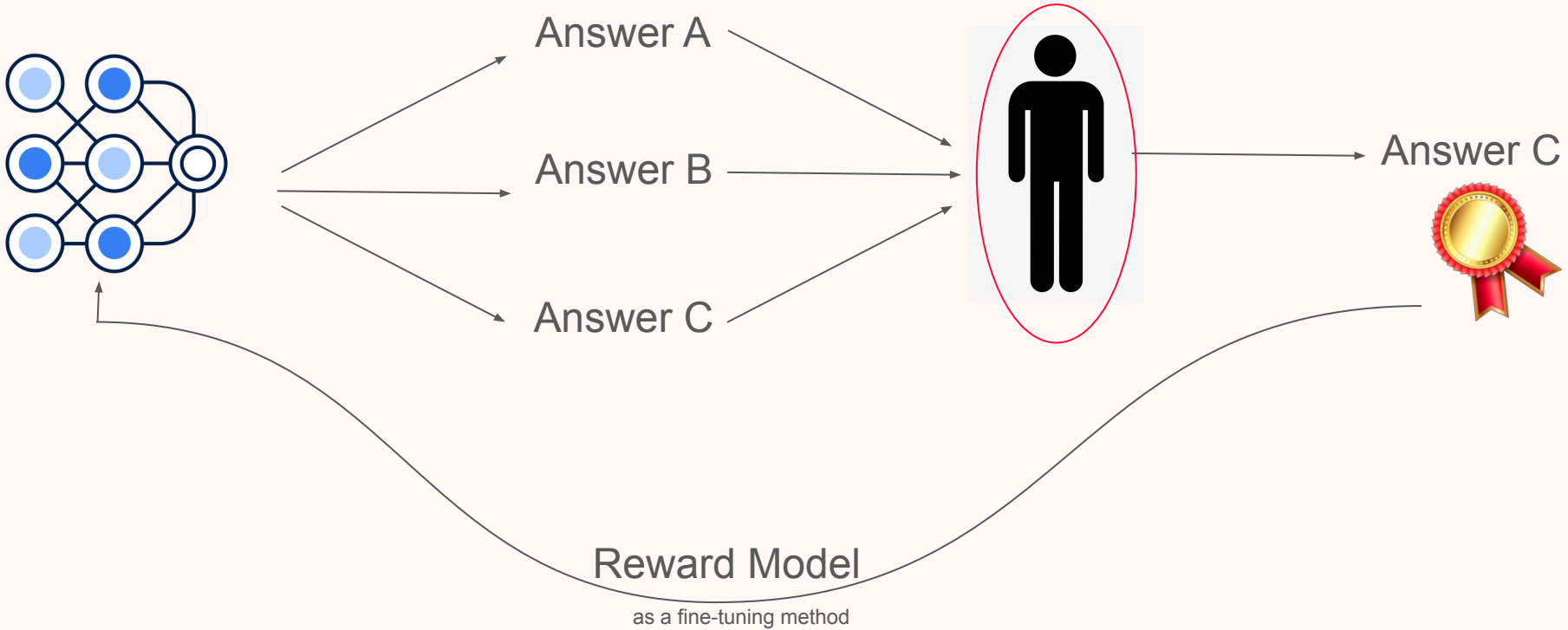


Reinforcement learning from human feedback (**RLHF**)



Reinforcement learning from human feedback (**RLHF**)

and its parametric limitation



Reinforcement learning from human feedback (**RLHF**)

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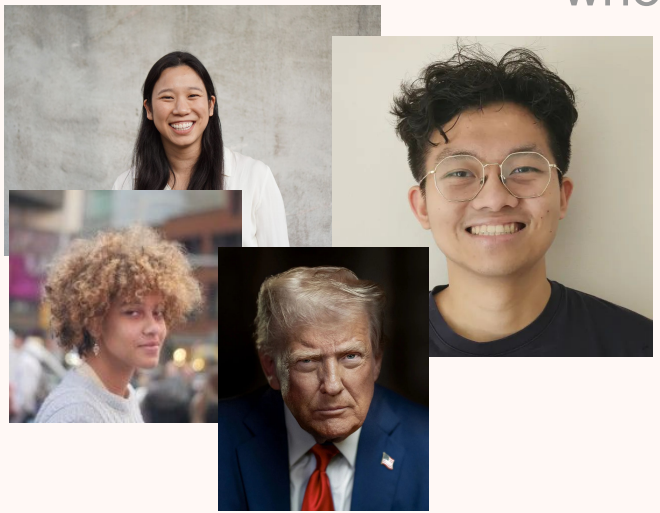
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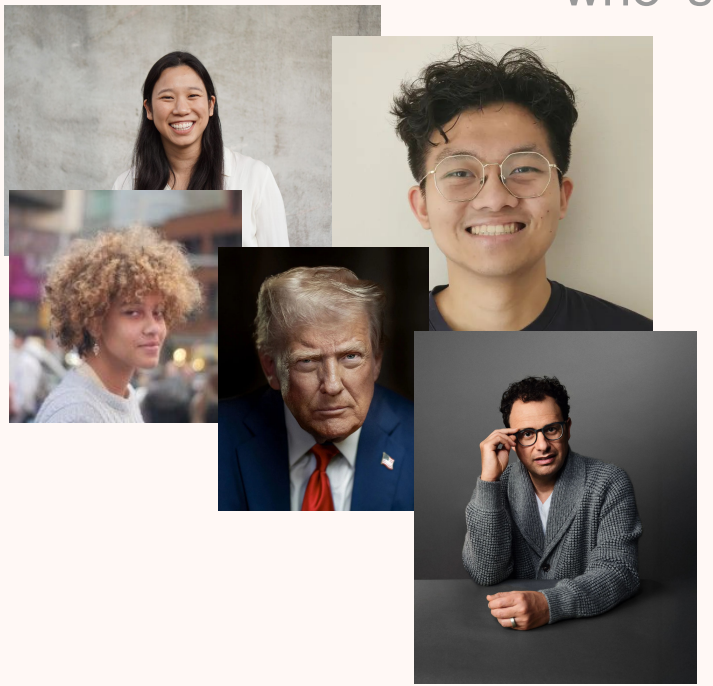
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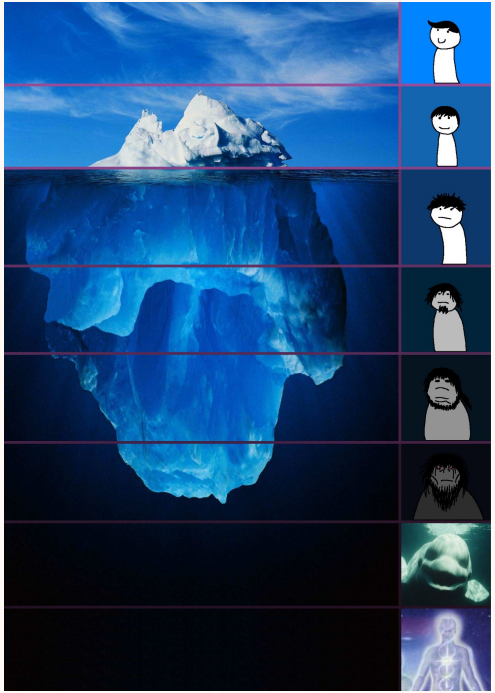
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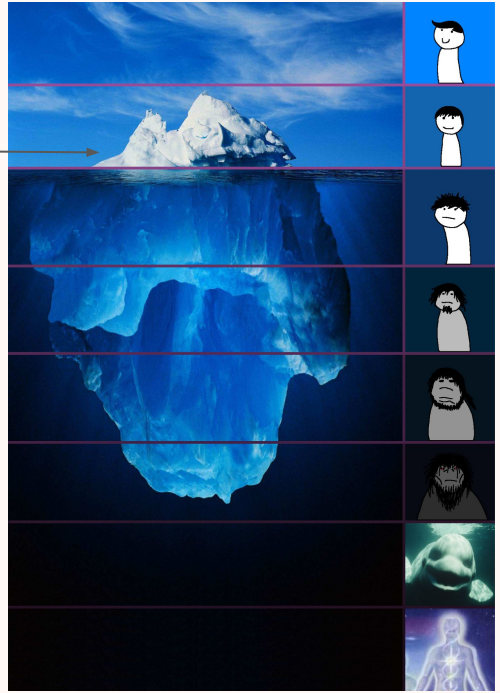
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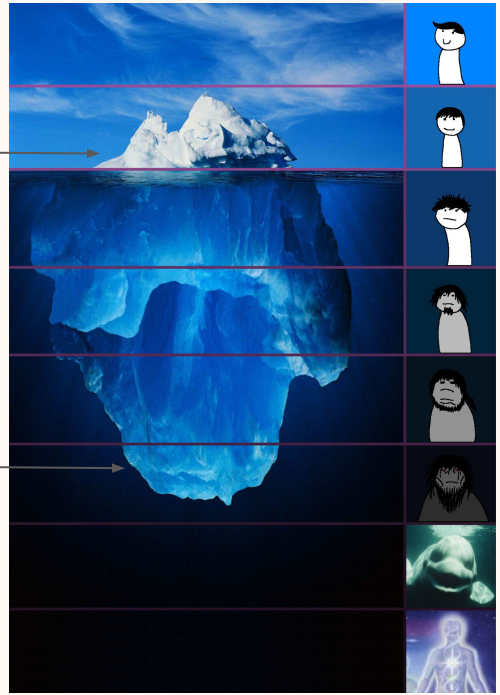


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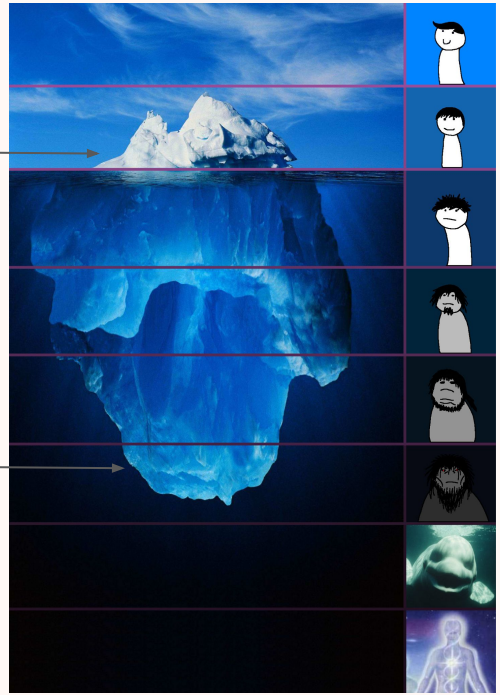


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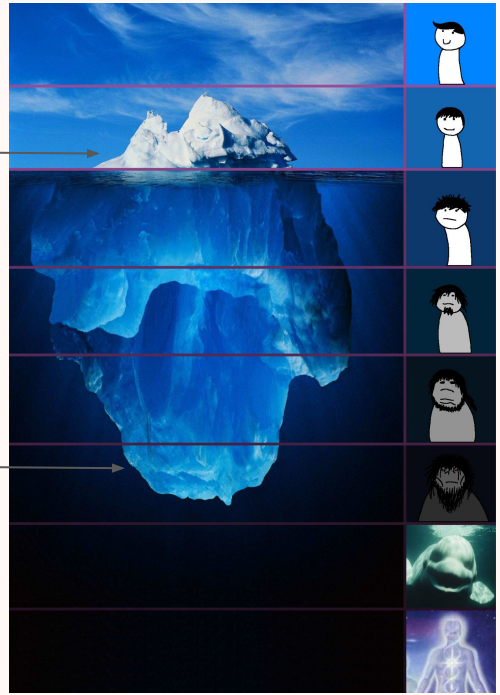
appearance of safety

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appearance of safety

Models actually learn to hide, rather than removing underlying capabilities or deep-seated risks.

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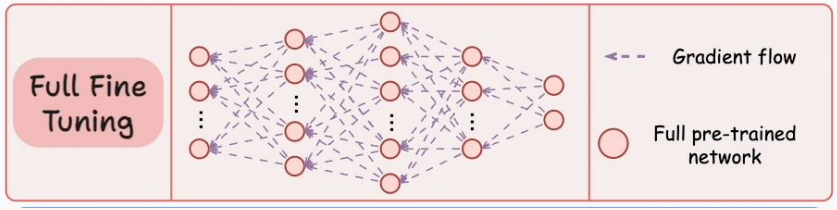
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Training on specific **prompt-and-response pairs** that explicitly model safe behavior:

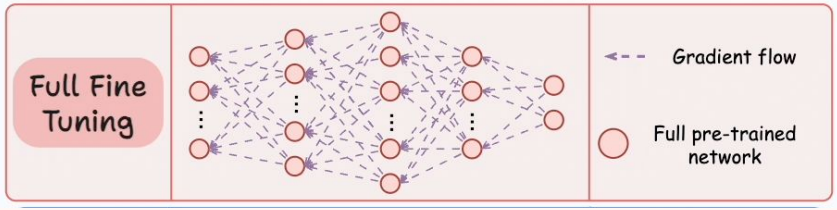
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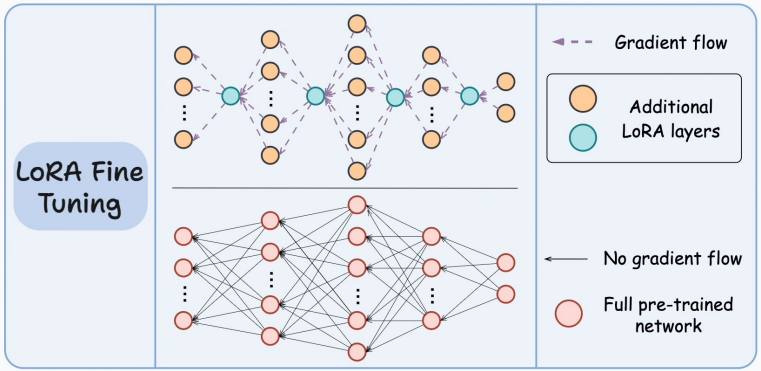
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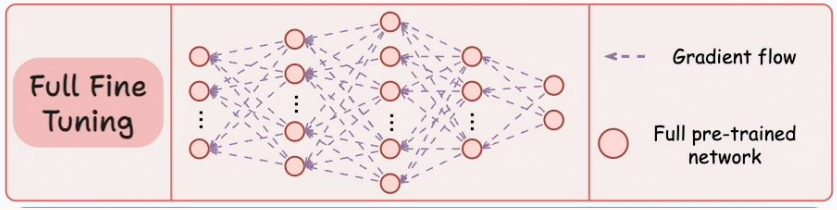
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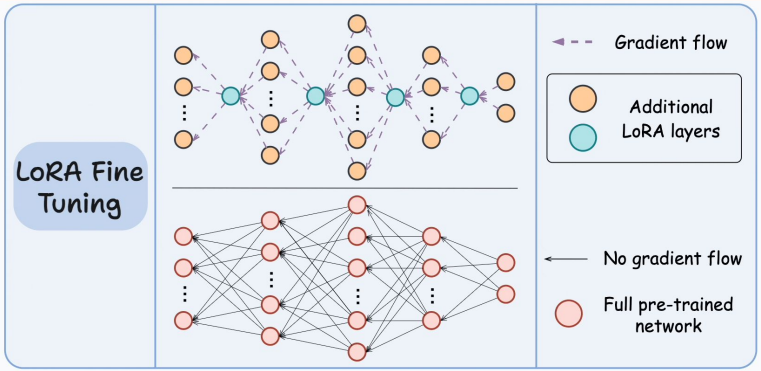




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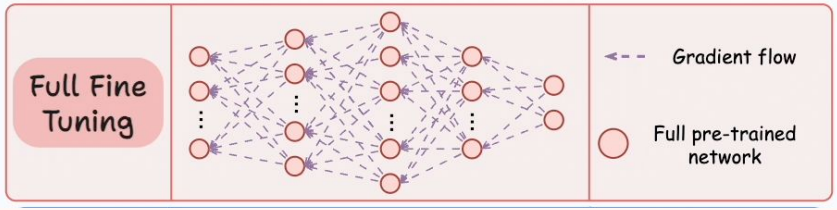


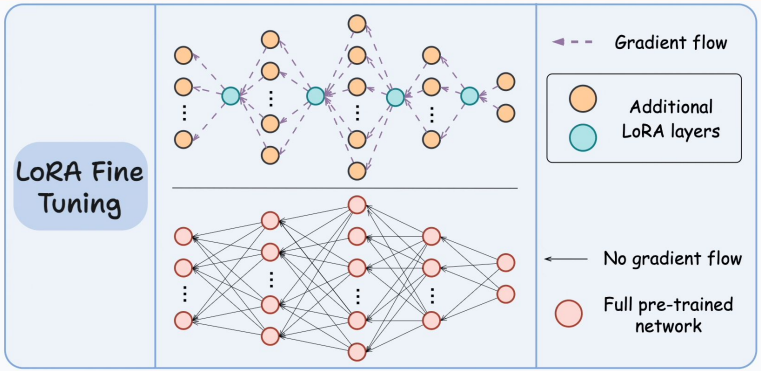
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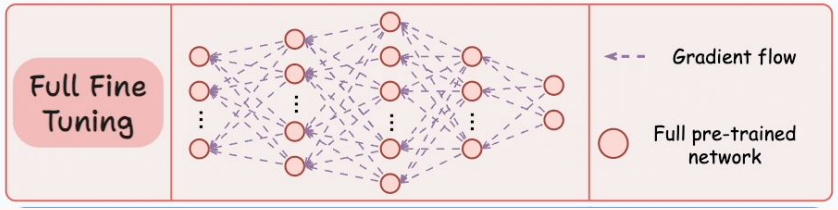


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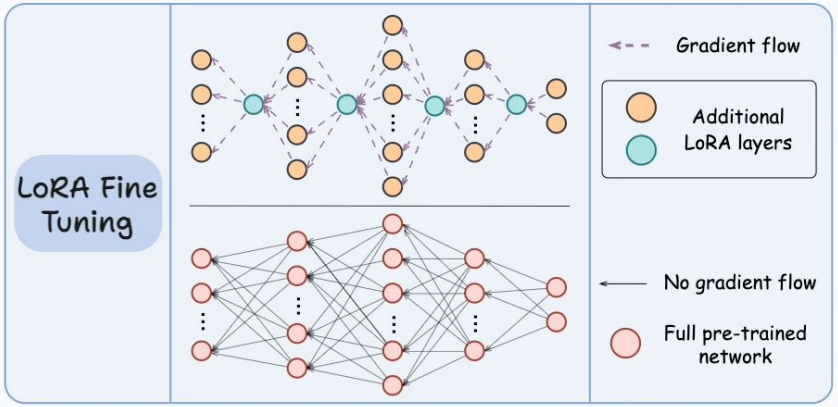
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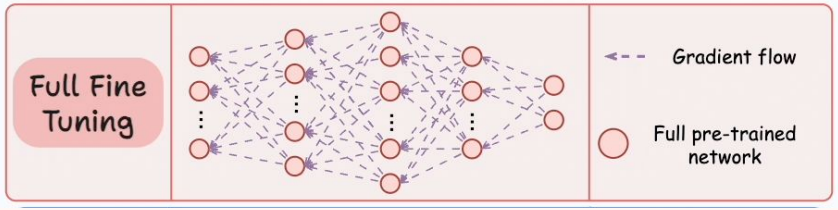


Compute Power



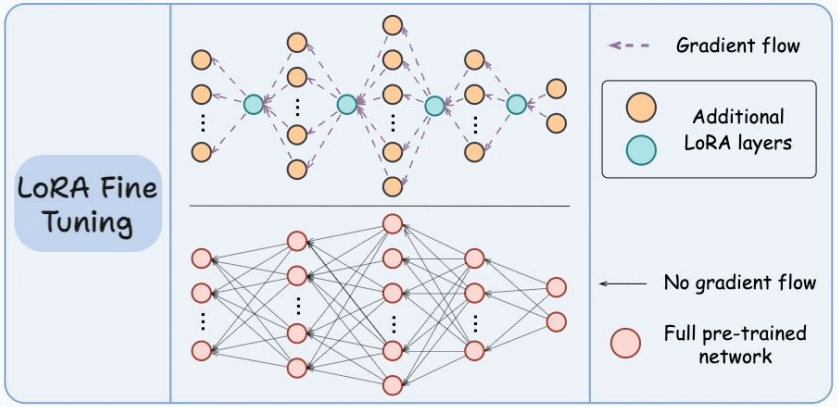
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Compute Power





Interacting with benign processes

Traditional Safety Alignment methods have **flaws**.

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Superficial alignment

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Superficial alignment



Expensive compute

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Causal Abstraction Framework

Geiger et al., 2023

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“formal mathematical and logical framework that precisely maps a low-level model (like a complex neural network) to a high-level model (like an intuitive human algorithm or causal graph) while preserving how the system responds to interventions.”

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Representation and space alignment intervention

- Concept Concentration (COCA)

How to avoid superficial alignment?

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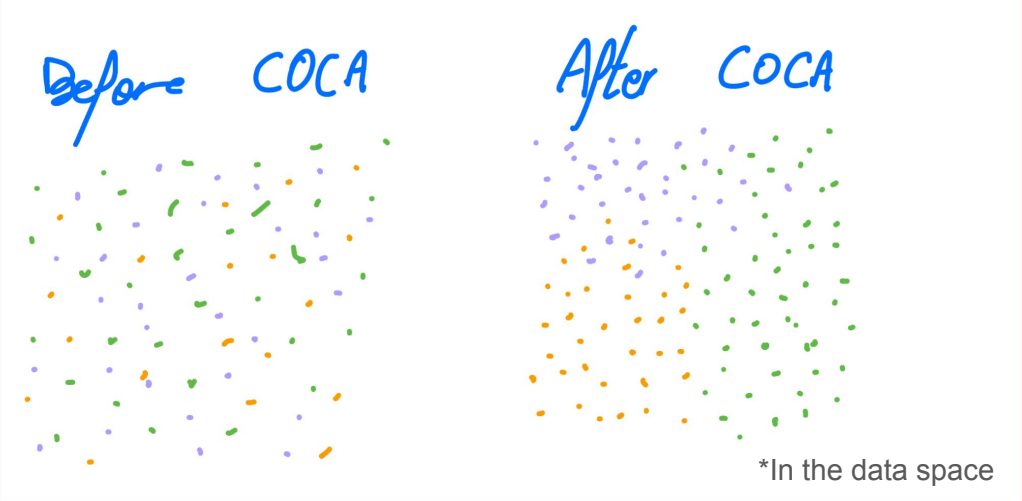
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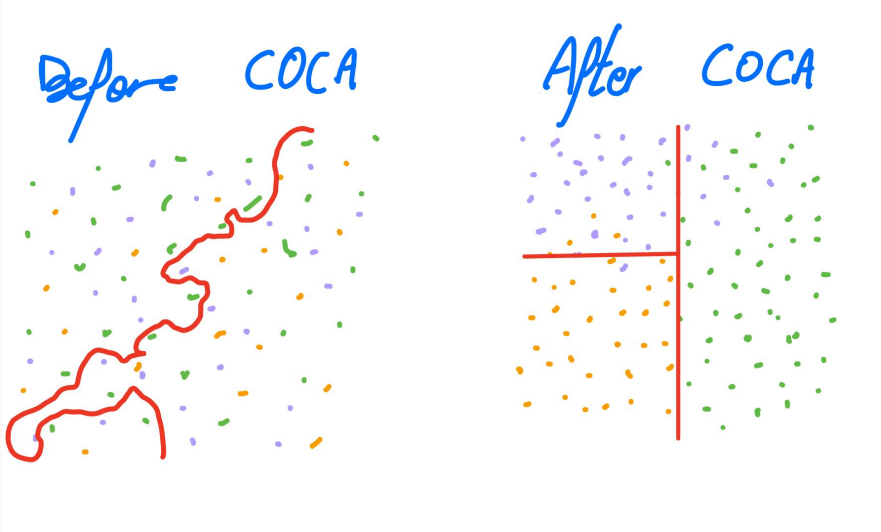
Geometry of the representation space is all you need.

COncept ConcentrAtion (COCA)

Act as a localized spatial interceptor rather than shifting the structural weights

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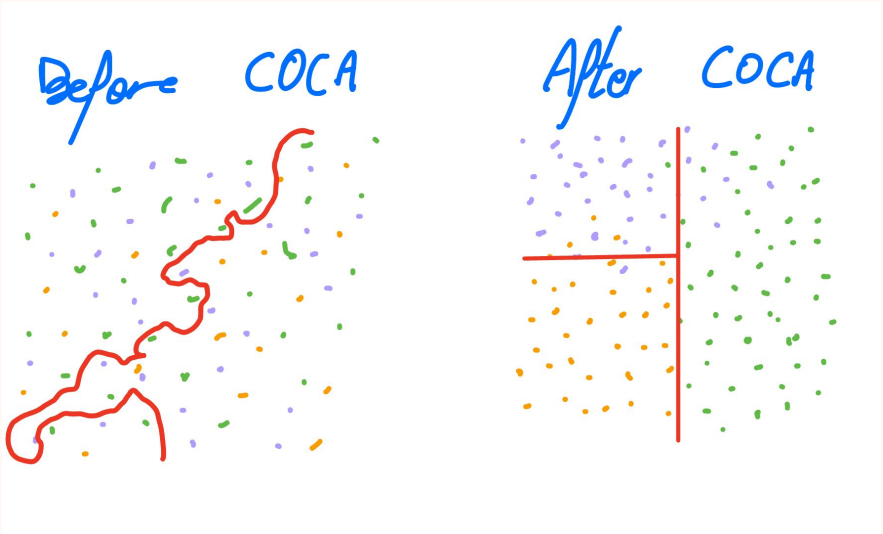


*In the primal space

Data space

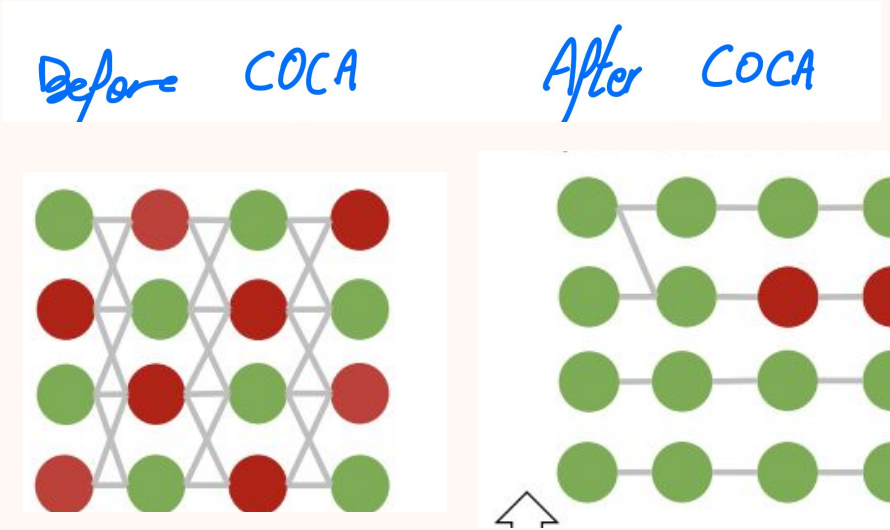
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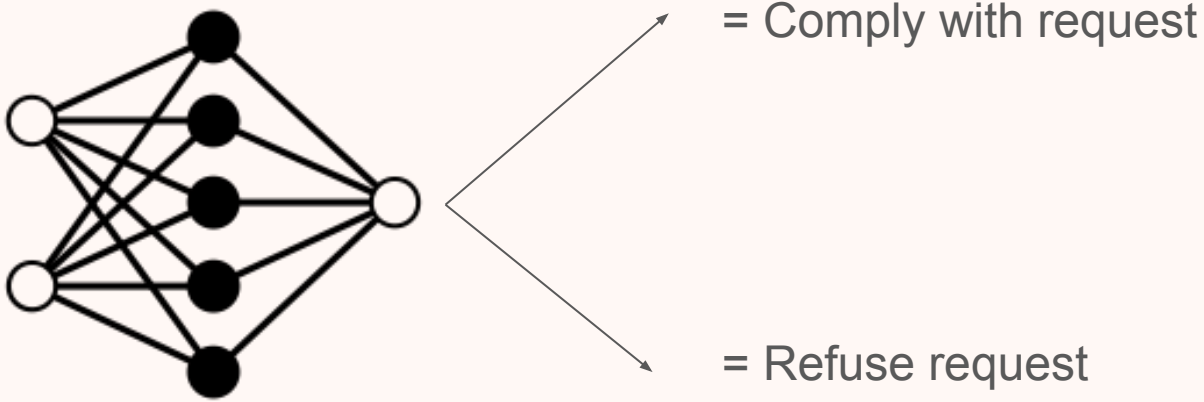
*In the dual space

Weights space

COCA : an **identify-then-respond** structure

Refactors safety training data to force the model to process harmful concepts through a structured and interpretable pipeline

- reshape the internal representation space
- simplifying the decision boundary between harmful and benign concepts



How to avoid superficial alignment?

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Representation and space alignment intervention

- Representational Similarity Analysis / Centered Kernel Alignment
  - Diagnostic tools that compare internal representations across different neural networks or human cognitive systems by evaluating second-order isomorphism (similarity structures)

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Representation and space alignment intervention

- Representational Similarity Analysis / Centered Kernel Alignment

- Disentangled Representation Learning
  - Frameworks that separate neural representations into independent, human-defined concepts using supervised contrastive learning with invariant risk minimization

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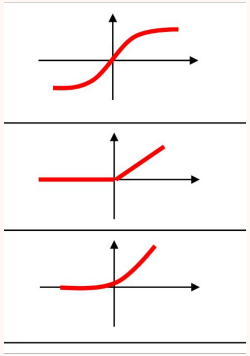
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Linear representation hypothesis

“concepts reside in linear subspaces of the activation space, allowing harmful concepts to be erased via linear transformations”



In non-linear settings, perfect concept erasure is impossible without distorting benign information

Instead of searching for ideal interventions in a complex space, simplify the space itself

